

EMMANUEL KONAN

Cybersecurity Engineer

✉ emmanuel.konan.contact@gmail.com | 🌐 kmanu225.github.io | [in](#) [emmanuel-konan](#) | [kmanu225](#)

Cybersecurity consultant specialized in IT, OT and IoT security, with experience in risk analysis, security audits, industrial systems and embedded security. I have worked across the energy, financial, public and IT/software sectors, combining technical expertise with cybersecurity governance and risk management.

RELEVANT PROFESSIONAL EXPERIENCE

Ornisec

2024-Present

Auditor & Cybersecurity Consultant - IT/OT/IoT

Cybersecurity consultant working across public/administrative organizations, financial institutions, energy companies and IT/software companies, covering risk management, technical audits, industrial cybersecurity and compliance.

- Lead EBIOS-RM risk analyses and facilitate workshops with business and technical stakeholders.
- Perform PASSI security audits covering architecture, configuration, source code, organizational and physical security.
- Conduct internal ISO/IEC 27001 audits, assessing controls, governance and ISMS effectiveness.
- Support clients on regulatory compliance, running NIS2 gap analyses and remediation roadmaps, and assessing AI Act obligations for AI-enabled systems.
- Define security requirements and conduct qualification and audits of industrial, embedded and connected equipment.
- Act as cybersecurity referent for OT/IoT equipment at Enedis, advising on PKI/HSM, PKCS#11, cryptographic mechanisms and asset mapping.
- Participate in the preparation and facilitation of cybersecurity crisis exercises, testing incident response, coordination and decision-making.
- Design and deliver cybersecurity training to client companies, covering technical, governance and awareness topics for both expert and non-expert audiences.
- Deliver cybersecurity engagements across administrative, financial, energy and IT/software environments.
- Integration of post-quantum cryptography (PQC) algorithms into environments with limited compute and storage resources.
- Led the firm's AI adoption initiative, identifying use cases and supporting secure integration into consulting workflows.

Enedis

Oct 2021 - Sep 2024

IT/OT Cybersecurity Engineer - Work-study

Sep 2023 - Sep 2024

Worked within a multidisciplinary team of cybersecurity experts on the security of embedded systems, smart metering and critical infrastructure.

- Assessed security risks and threat models related to security modules embedded in industrial systems.
- Analyzed defense-in-depth architectures, hardware security (CPU, memory, buses) and potential vulnerabilities.
- Conducted research on cryptographic mechanisms, SSH protocols (RFC 4250-4254), PLC and ISO/IEC 7816 smart-card standards.
- Performed vulnerability research and penetration-testing activities on embedded environments.
- Modified the OpenSSH client in C for custom security testing and information gathering.
- Developed Python and Bash scripts to automate security testing and analysis.

Non-Technical Losses Project Manager - Work-study

Oct 2021 - Aug 2023

Worked within Enedis' B2B division on initiatives aimed at reducing non-technical losses and electricity fraud, combining data analysis, project management and operational coordination.

- Performed data mining and consumption analysis to identify potential fraud and abnormal electricity usage.
- Automated monthly reporting and KPIs using VBA, improving performance monitoring.
- Managed a national project introducing a new feature within the electricity distribution network.
- Conducted security and compliance assessments of internal business tools.
- Supported the development of Enedis' B2B strategic plan and internal awareness initiatives.
- Completed a field immersion with operational technicians, gaining practical knowledge of electricity distribution, non-technical losses and the French energy market.

EDUCATION

CentraleSupélec

Engineering Degree - Computer Science & Cybersecurity

2021-2024

GPA 4.23/5.0. General engineering curriculum spanning computer science, cybersecurity, electronics, data/AI and management.

Université de Rennes

Advanced Hardware Security - Master 2 coursework

2023-2024

Coursework in embedded-system attacks, including side channels, glitching and JTAG.

Saint Aubin La Salle

Preparatory Classes - PTSI/PT

2019-2021

Mathematics, physics and engineering sciences. Ranked first of the class, highest honors.

Lycée Classique d'Abidjan, Côte d'Ivoire

Baccalauréat Série C - Mathematics & Physics/Chemistry

2016-2019

Graduated with honors (Mention Bien). Ranked first of the class.

PUBLICATIONS & STANDARDS CONTRIBUTIONS

1. **EAP-PSK-256 - Cryptographic protocol contribution** IETF Internet-Draft, 2026. Coauthor of an individual Internet-Draft targeting quantum-resistant authentication for constrained devices, using pre-shared keys, AES-256 and NIST SP 800-108 key derivation. Work in progress; not an RFC or an IETF-endorsed standard. datatracker.ietf.org/doc/draft-eap-psk-256

SKILLS & INTERESTS

Certifications: ISO 27001 Lead Auditor (2026); eJPT (2025)

Languages: French; English - TOEIC 910/990, Linguaskill C1

Core strengths: Leadership & motivation; Proactivity & initiative; Solution-oriented mindset; Problem-solving; Adaptability; Time management & organization

Community:

- Business Development Lead & IT Consultant, [African Engineers Network](#) (2024-Present)
- Digital Club Member, [Elit Club](#) (2025-Present)
- Private Tutor, Superprof (2022-2024)
- Treasurer & IT Support, [Rézo Metz-Rennes Fédérés](#) (2021-2024)